MIL-W-60197(MU)
AMENDMENT 4
28 July 1969
SUPERSEDING
AMENDMENT 3
12 July 1968

MILITARY SPECIFICATION

WRITE AMPLIFIER: 10526585

This amendment forms a part of Military Specification MIL-W-60197(MU), dated 24 February 1965.

Paragraph 3.1: Change Drawing Number "F10526858" to "F10526585".

Paragraph 3.4.1: Delete Pin "32B".

Paragraph 3.4.1.1(g): Change "2 Milliamperes" to "4 Milliamperes".

* Delete "Figure 1" and substitute new "Figure 1 attached".

Following paragraph 4.1: Add paragraphs 4.1.1, 4.1.2, 4.1.2.1 and 4.1.2.2.

* "4.1.1 General provisions.- The component inspection requirements of MIL-F-13926 form a part of the Quality Assurance Provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.1.2 First Article (Initial Production) Approval.- The requirement for first article approval and the responsibility (Government or contractor) for first article testing shall be as specified in the contract. The sample for first article approval tests shall consist of three (3) units manufactured in the same manner using the same materials, equipment, processes, and procedures as used in regular production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production.

4.1.2.1 Government Testing.- When the Government is responsible for conducting first article approval tests, the contractor, prior to submitting the sample to the Government, shall inspect the sample to insure that it conforms to all the requirements of the contract and submit a record of this inspection with the sample, including certificates of conformance for materials.

4.1.2.2 Contractor Testing.- When the contractor is responsible for conducting first article approval tests, the sample shall be inspected and a record of this inspection, including certificates of conformance for materials shall be submitted to the Government for approval. The Government reserves the right to witness the contractor's inspection."

FSC 1220

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* Paragraph 4.2.3: Add the following sentence: "No lots shall be accepted until control samples have been satisfactorily tested".

* Paragraph 4.3.2: Delete first sentence and substitute:

"The units shall be selected at random by a government representative as control samples from each three (3) months production or portion thereof as follows:

3 Units per first 100 produced
5 units per 200
7 units per 300
9 units per 400
10 units per 500
12 units per 700
15 units per 1000
and 2 units for each 100 items produced in excess of 1000".

* Paragraph 4.4.1.1: Items 1 through 8: Delete "from dc to 500" and substitute "from dc to 20 kc". Delete items 9 and 10. Item 11, Reset voltage: Change "minus 25 $\pm$ 2 volts" to "minus "18 $\pm$ 2 volts."

Paragraph 4.5.1.1, Line 5 after "vibration": Delete the letter "c".

Paragraph 4.5.1.3.2: Delete in its entirety and substitute:

"4.5.1.3.2 Operating.- This test shall be conducted in accordance with the humidity test defined in Procedure III of MIL-F-13926, except that the humidity shall be maintained at 92 to 97 percent for the 24-hour period, and the test shall cease at the end of that period (Method 106 of MIL-STD-202 shall be excluded). Throughout the test period, the board shall be capable of performing the test of 4.5.3.3. In order to demonstrate this capability, the tests of 4.5.3.3 shall be performed a minimum of once each 6 hours of the 24-hour period. The final test shall be conducted during the final portion of the 24-hour period. Between tests, the board shall be electrically energized as described in 4.5.2.1."

Paragraph 4.5.1.4: Delete in its entirety and substitute:

"4.5.1.4 Salt spray.- This test shall be conducted in accordance with the salt spray test defined in Method 101, Condition B of MIL-STD-202 using a 5 percent salt solution. At the end of the exposure period, salt deposits shall be removed from the board by a gentle wash, not exceeding 30 minutes, in running water not warmer than 100°F, using a soft-hair brush or plastic-bristle brush. The board shall then be air dried in a controlled environment at 72°F $\pm$ 5°F and a relative humidity of 30 to 50 percent for a period of 24 hours. The board shall be inspected visually for any physical damage, then tested in accordance with 4.5.3.3. The board shall subsequently be inspected as specified in 4.5.1.5, prior to determining compliance with 3.4 as required by 3.3.6."

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Paragraph 4.5.2.1: Change Pin "32B" to "23A".

Paragraph 4.5.2.1.1: Delete Pin "32B".

* Paragraph 4.5.3.1: Change "set voltage" to "reset voltage".

Item (a): Shall read, "The voltage from Pin 17A to Pin 52B shall be more positive than minus 0.3 volts".

Item (c): Change to read _ _ _ _Pin 11B to Pin 33A shall _ _ _ _.

Item (d): Shall read, "The voltage at Pin 1B to Pin 33A shall be equal to or less than plus 12 volts".

Item (e): Change to read _ _ _ _Pin 25B to Pin 33A shall _ _ _ _.

Item (f): Change to read _ _ _ _Pin 22B to Pin 33A shall _ _ _ _.

Item (g): Change "64 $\pm$ 2" to "64 $\pm$ 4".

* Paragraph 4.5.3.2: Change "Pin 13B" to "Pin 15A".

Item (a) shall read, "_ _ _ _shall be more positive than minus 0.3 volts".

Item (c) shall read "_ _ _ _Pin 22B to 33A shall be_ _ _ _."

Item (d) shall read "_ _ _ _Pin 18B to Pin 33A shall be equal to or less than plus 12 volts."

Item (e) shall read "_ _ _ _Pin 4B to Pin 33A shall be _ _ _ _."

Item (f) shall read "_ _ _ _Pin 11B to Pin 33A shall be_ _ _ _."

Item (g): Change "64 $\pm$ 2" to "64 $\pm$ 4".

* Paragraph 4.5.3.3.1.1: Change "(Item 12B)" to "(item 13B)". Change "0.5 $\pm$ 0.75 volt".

* Paragraph 4.5.3.3.1.2: Change "(item 12C)" to "(item 13C)".

* Paragraph 4.5.3.3.2: Change "(item 12A)" to "(item 13A)".

* Paragraph 4.5.3.3.2.1: Change "(item 11)" to "(item 13A)". Delete the third sentence.

* Paragraph 4.5.4: Add "36A" after Pin "34B" and "29A" after "52B".

Table V: Delete Pins "29A and 36A"

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Change "32B" to "23A".

Add pins "5A and 24A" to Write Amplifier 1 column and Pins "40B and 62B" to Write Amplifier 2 column.

Paragraph 4.5.5.1: Change Pin "36B" to "63B".

Paragraph 4.5.6.1.1: Delete in its entirety and substitute:

..."4.5.6.1.1 Forward voltage drop.- The forward voltage drop shall be tested by the following procedure to determine compliance with 3.4.4.1.

(a) Connect the minus 18 volt power supply to Pin 50B. The voltage at the pin in Column A with respect to the pin in Column B shall be more positive than minus 0.6 volt. Connect the power supply ground return in turn to the pin in Column B.

Column A	Column B
10B	5A
10B	14A
19B	24A
19B	14A

After the last measurement is complete, remove the power supply ground return from Pin 14A. Remove the minus 18 volt power supply from Pin 50B.

(b) Connect the minus 18 volt power supply to Pin 49B. The voltage at the pin in Column A with respect to the pin in Column B shall be as shown in Column C. Connect the power supply ground return in turn to the pin in Column B.

Column A	Column B	Column C
8B	2B	Minus 0.6
8B	33B	Minus 0.8
20B	23B	Minus 0.6
20B	26B	Minus 0.8
20B	14A	Minus 0.6

After the last measurement is complete, remove the ground return from 14A. Remove the minus 18 volt power supply from Pin 49B.

(c) Connect the minus 18 volt power supply thru a 68 K ohm 1/4 W 5% resistor to Pin 14A. Measure the voltage at Pin 14A with respect to the grounded pin. The voltage shall be more positive than minus 0.6 volts. Connect the power supply ground return in turn to Pin 20A and Pin 22A. Upon completion of the forward voltage drop tests remove the minus 18 volt power supply from Pin 14A and all leads from the module."

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Paragraph 4.5.6.1.2: Delete in its entirety and substitute:

"4.5.6.1.2 Leakage current.- Leakage current shall be tested by the following procedures to determine compliance with 3.4.4.2.

(a) Connect the plus 15 volt power supply to Pin 49B. Connect the power supply ground return to pins 33B and 26B. The voltage from Pin 49B to Pins 8B and 20B shall be less than 0.05 volt. Remove the power supply ground return from Pins 33B and 26B.

(b) Connect the plus 15 volt power supply ground return to Pins 2B and 23B. The voltage from Pin 49B to Pins 8B and 20B shall be less than 1.7 volts. Remove the power supply ground return from Pins 2B and 23B.

(c) Connect the plus 15 volt power supply ground return to Pin 14A. The voltage from Pin 49B to Pins 8B and 20B shall be less than 1.7 volts. Remove the plus 15 volt power supply from Pin 49B and the ground return from Pin 14A.

(d) Connect the plus 15 volt power supply to Pin 50B. Connect the power supply ground return to Pin 5A and 24A. The voltage from Pin 50B to Pins 10B and 19B shall be less than 1.18 volts. Remove the power supply ground return from Pins 5A and 24A.

(e) Connect the plus 15 volt power supply ground return to Pin 14A. The voltage from Pin 50B to Pins 10B and 19B shall be less than 1.18 volts. Remove the plus 15 volt power supply from Pin 50B and the ground return from Pin 14A.

(f) Connect the plus 15 volt power supply thru a 68 K ohm 1/4 W ± 5 percent resistor to Pin 14A. Connect the power supply ground return to Pin 20A. The voltage from the plus 15 volt power supply to Pin 14A shall be less than 1.7 volts. Move the plus 15 volt power supply ground return from Pin 20A to Pin 22A. The voltage from the plus 15 volt power supply to Pin 14A shall be less than 1.7 volts. Remove the power supply ground return from Pin 22A and the connection from Pin 14A."

Figure 3: Delete and substitute new Figure 3.

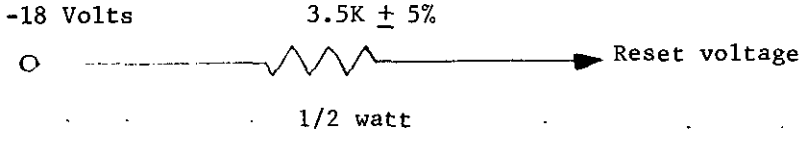


FIGURE 3

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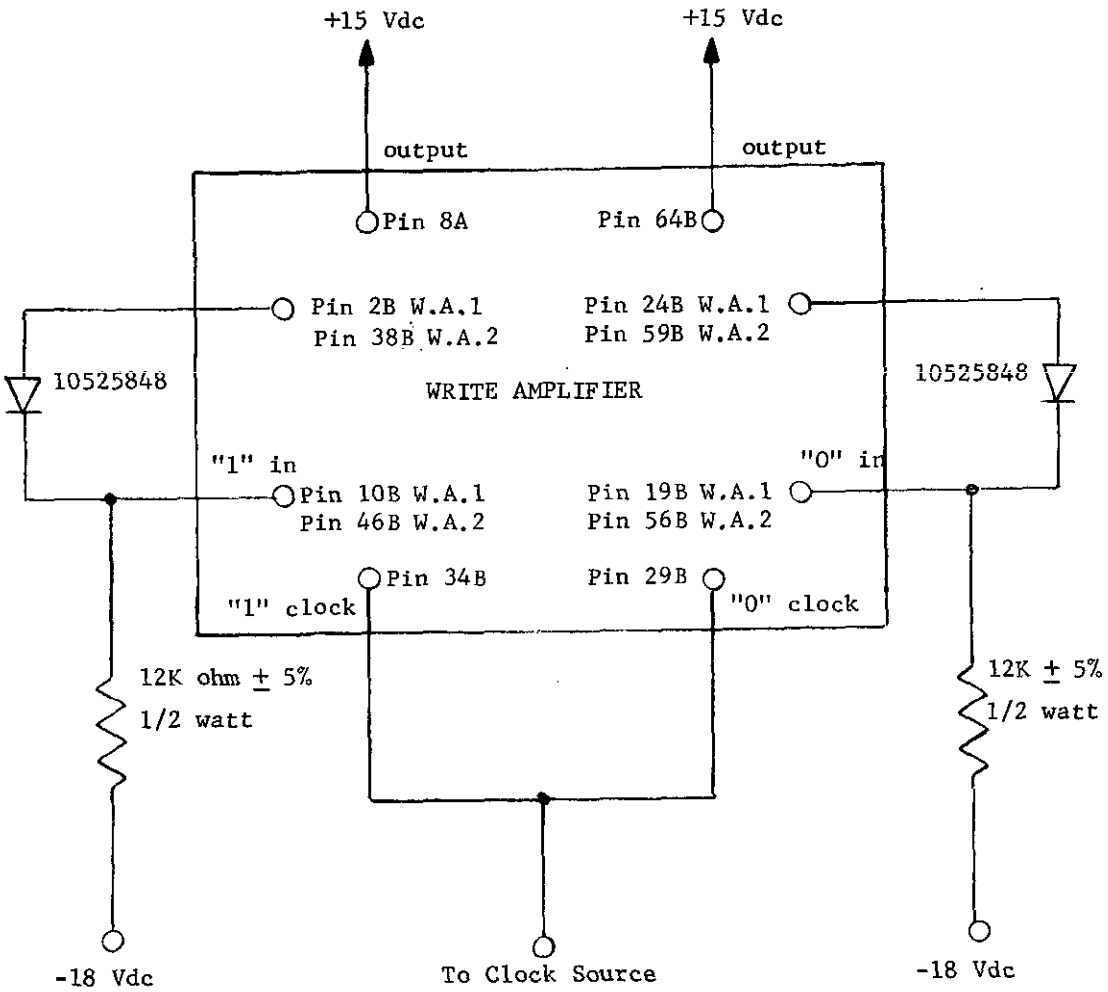


FIGURE 1 - Write Amplifier Self-Gating System

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